

Vectors

GCSE Maths · Worked Solutions

20 questions · 62 marks total

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Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

Write the translation '3 units right and 2 units down' as a column vector.

- 1 The top number is the horizontal move (right is positive).
- 2 The bottom number is the vertical move (down is negative).

Answer: $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$

2 [2 marks]

Work out $\begin{pmatrix} 2 \\ 5 \end{pmatrix} + \begin{pmatrix} 3 \\ -1 \end{pmatrix}$

- 1 Add the top numbers and add the bottom numbers separately.
- 2 Top: $2 + 3 = 5$
- 3 Bottom: $5 + (-1) = 4$

Answer: $\begin{pmatrix} 5 \\ 4 \end{pmatrix}$

3 [2 marks]

Work out $\begin{pmatrix} 6 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 5 \end{pmatrix}$

- 1 Subtract the top numbers and the bottom numbers separately.
- 2 Top: $6 - 1 = 5$
- 3 Bottom: $2 - 5 = -3$

Answer: $\begin{pmatrix} 5 \\ -3 \end{pmatrix}$

4 [2 marks]

Work out $3 \begin{pmatrix} 2 \\ -4 \end{pmatrix}$

- 1 Multiply both numbers in the vector by 3.
- 2 $3 \times 2 = 6$ and $3 \times (-4) = -12$

Answer: $\begin{pmatrix} 6 \\ -12 \end{pmatrix}$

5

[3 marks]

$$\mathbf{a} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \text{ and } \mathbf{b} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$$

Work out $\mathbf{a} + 2\mathbf{b}$

1 First work out $2\mathbf{b}$: $2 \times (-2) = -4$ and $2 \times 4 = 8$, giving $\begin{pmatrix} -4 \\ 8 \end{pmatrix}$

2 Now add $\mathbf{a}$: top $3 + (-4) = -1$

3 Bottom: $1 + 8 = 9$

Answer: $\begin{pmatrix} -1 \\ 9 \end{pmatrix}$

6

[3 marks]

$$\mathbf{a} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \text{ and } \mathbf{b} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$$

Work out $2\mathbf{a} - \mathbf{b}$

1 $2\mathbf{a} = \begin{pmatrix} 6 \\ 2 \end{pmatrix}$

2 Subtracting $\mathbf{b}$ means subtracting -2 and 4 .

3 Top: $6 - (-2) = 8$

4 Bottom: $2 - 4 = -2$

Answer: $\begin{pmatrix} 8 \\ -2 \end{pmatrix}$

7

[3 marks]

A is the point $(1, 2)$. A is translated by $\begin{pmatrix} 4 \\ -3 \end{pmatrix}$ to give B.

(a) Write down the coordinates of B.

(b) Write down the vector that would translate B back to A.

1 (a) x : $1 + 4 = 5$ and y : $2 + (-3) = -1$, so B is $(5, -1)$

2 (b) Going back undoes the move, so change the sign of both numbers.

3 The vector is $\begin{pmatrix} -4 \\ 3 \end{pmatrix}$

Answer: (a) $(5, -1)$ (b) $\begin{pmatrix} -4 \\ 3 \end{pmatrix}$

8

[3 marks]

Work out the magnitude (length) of

(a) $\begin{pmatrix} 6 \\ 8 \end{pmatrix}$ (b) $\begin{pmatrix} 5 \\ 12 \end{pmatrix}$

1 Use Pythagoras: magnitude $= \sqrt{x^2 + y^2}$

2 (a) $\sqrt{36 + 64} = \sqrt{100} = 10$

3 (b) $\sqrt{25 + 144} = \sqrt{169} = 13$

Answer: (a) 10 (b) 13

9

[4 marks]

The vector from O to A is $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ and the vector from O to B is $\begin{pmatrix} 5 \\ 7 \end{pmatrix}$

(a) Work out the vector from A to B.

(b) Work out the length of AB.

1 (a) To go from A to B, travel back to O then out to B.

2 $AB = -OA + OB = OB - OA$

3 Top: $5 - 2 = 3$ and bottom: $7 - 3 = 4$, giving $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$

4 (b) Length = $\sqrt{(3^2 + 4^2)} = \sqrt{25} = 5$

Answer: (a) $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ (b) 5

10

[4 marks]

(a) Show that $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$ and $\begin{pmatrix} 6 \\ 9 \end{pmatrix}$ are parallel.

(b) Write $\begin{pmatrix} 10 \\ 15 \end{pmatrix}$ as a multiple of $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$

1 Two vectors are parallel when one is a multiple of the other.

2 (a) $6 \div 4 = 1.5$ and $9 \div 6 = 1.5$

3 Both numbers scale by the same factor, so $\begin{pmatrix} 6 \\ 9 \end{pmatrix} = 1.5 \begin{pmatrix} 4 \\ 6 \end{pmatrix}$ and the vectors are parallel.

4 (b) $10 \div 2 = 5$ and $15 \div 3 = 5$

5 So $\begin{pmatrix} 10 \\ 15 \end{pmatrix} = 5 \begin{pmatrix} 2 \\ 3 \end{pmatrix}$

Answer: (a) the second is 1.5 times the first (b) $5 \begin{pmatrix} 2 \\ 3 \end{pmatrix}$

Section B · Higher

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1

[2 marks]

$\mathbf{a} = \begin{pmatrix} 4 \\ -1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$

Work out $3\mathbf{a} + 2\mathbf{b}$

1 $3\mathbf{a} = \begin{pmatrix} 12 \\ -3 \end{pmatrix}$ and $2\mathbf{b} = \begin{pmatrix} -4 \\ 6 \end{pmatrix}$

2 Top: $12 + (-4) = 8$ and bottom: $-3 + 6 = 3$

Answer: $\begin{pmatrix} 8 \\ 3 \end{pmatrix}$

2

[2 marks]

Work out the magnitude of $\begin{pmatrix} -7 \\ 24 \end{pmatrix}$

1 Magnitude = $\sqrt{((-7)^2 + 24^2)}$

2 = $\sqrt{(49 + 576)} = \sqrt{625} = 25$

3 Squaring removes the minus sign, so a magnitude is never negative.

Answer: 25

3

[3 marks]

In triangle OAB, the vector from O to A is **a** and the vector from O to B is **b**.

Write the vector from A to B in terms of **a** and **b**.

- 1 There is no direct information about AB, so build a route through O.
- 2 $AB = AO + OB$
- 3 AO is the reverse of OA, so $AO = -a$

Answer: $b - a$

4

[3 marks]

In triangle OAB, $OA = a$ and $OB = b$. M is the midpoint of AB.

Write the vector from O to M in terms of **a** and **b**.

- 1 Travel from O to A, then half way along AB.
- 2 $AB = b - a$, so $AM = \frac{1}{2}(b - a)$
- 3 $OM = a + \frac{1}{2}(b - a) = \frac{1}{2}a + \frac{1}{2}b$

Answer: $\frac{1}{2}(a + b)$

5

[4 marks]

In triangle OAB, $OA = a$ and $OB = b$. P lies on AB so that $AP : PB = 1 : 3$

Write the vector from O to P in terms of **a** and **b**.

- 1 $AP : PB = 1 : 3$ means AB is split into 4 equal parts, and P is 1 part along.
- 2 So $AP = \frac{1}{4}AB = \frac{1}{4}(b - a)$
- 3 $OP = OA + AP = a + \frac{1}{4}(b - a)$
- 4 Expand: $a - \frac{1}{4}a + \frac{1}{4}b$
- 5 $= \frac{3}{4}a + \frac{1}{4}b$

Answer: $\frac{3}{4}a + \frac{1}{4}b$

6

[4 marks]

$OC = 3a + 2b$ and $OD = 6a + 4b$.

Show that O, C and D lie on the same straight line.

- 1 Look for a multiple linking the two vectors.
- 2 $6a + 4b = 2(3a + 2b)$
- 3 So $OD = 2 \times OC$.
- 4 Because OD is a multiple of OC, the two vectors are parallel.
- 5 They also both start at O, so the three points must lie on one straight line (with D twice as far from O as C).

Answer: $OD = 2 OC$, so the vectors are parallel and share the point O, making O, C and D collinear

7

[4 marks]

$OA = 2\mathbf{a} + \mathbf{b}$, $OB = 4\mathbf{a} + 3\mathbf{b}$ and $OC = 7\mathbf{a} + 6\mathbf{b}$.

Show that A, B and C lie on the same straight line.

- 1 Find the vectors between the points, not just from O.
- 2 $AB = OB - OA = (4\mathbf{a} + 3\mathbf{b}) - (2\mathbf{a} + \mathbf{b}) = 2\mathbf{a} + 2\mathbf{b}$
- 3 $BC = OC - OB = (7\mathbf{a} + 6\mathbf{b}) - (4\mathbf{a} + 3\mathbf{b}) = 3\mathbf{a} + 3\mathbf{b}$
- 4 $BC = 1.5(2\mathbf{a} + 2\mathbf{b}) = 1.5 AB$, so AB and BC are parallel.
- 5 They share the point B, so A, B and C are collinear.

Answer: $AB = 2\mathbf{a} + 2\mathbf{b}$ and $BC = 1.5 AB$, so the points are collinear

8

[4 marks]

OACB is a parallelogram with $OA = \mathbf{a}$ and $OB = \mathbf{b}$.

M is the midpoint of AC.

Write the vector from O to M in terms of $\mathbf{a}$ and $\mathbf{b}$.

- 1 In parallelogram OACB, the side AC is parallel and equal to OB.
- 2 So $AC = \mathbf{b}$
- 3 M is the midpoint of AC, so $AM = \frac{1}{2}\mathbf{b}$
- 4 $OM = OA + AM = \mathbf{a} + \frac{1}{2}\mathbf{b}$

Answer: $\mathbf{a} + \frac{1}{2}\mathbf{b}$

9

[4 marks]

In triangle OAB, $OA = \mathbf{a}$ and $OB = \mathbf{b}$. P lies on AB with $AP : PB = 2 : 3$

Show that $OP = \frac{1}{5}(3\mathbf{a} + 2\mathbf{b})$

- 1 AB is divided into $2 + 3 = 5$ equal parts and P is 2 parts along from A.
- 2 $AP = \frac{2}{5}(\mathbf{b} - \mathbf{a})$
- 3 $OP = \mathbf{a} + \frac{2}{5}(\mathbf{b} - \mathbf{a})$
- 4 $= \mathbf{a} - \frac{2}{5}\mathbf{a} + \frac{2}{5}\mathbf{b} = \frac{3}{5}\mathbf{a} + \frac{2}{5}\mathbf{b}$
- 5 Taking out the factor $\frac{1}{5}$ gives $\frac{1}{5}(3\mathbf{a} + 2\mathbf{b})$ as required.

Answer: $OP = \frac{3}{5}\mathbf{a} + \frac{2}{5}\mathbf{b} = \frac{1}{5}(3\mathbf{a} + 2\mathbf{b})$

OABC is a parallelogram with $OA = \mathbf{a}$ and $OC = \mathbf{c}$.

M is the midpoint of AB and N is the midpoint of BC.

(a) Write OM in terms of $\mathbf{a}$ and $\mathbf{c}$.

(b) Write ON in terms of $\mathbf{a}$ and $\mathbf{c}$.

(c) Show that MN is parallel to AC, and state the ratio of their lengths.

1 (a) In the parallelogram, AB is parallel and equal to OC, so $AB = \mathbf{c}$.

2 M is halfway along AB, so $OM = \mathbf{a} + \frac{1}{2}\mathbf{c}$

3 (b) Likewise CB is parallel and equal to OA, so $CB = \mathbf{a}$.

4 N is halfway along CB, so $ON = \mathbf{c} + \frac{1}{2}\mathbf{a}$

5 (c) $MN = ON - OM = (\mathbf{c} + \frac{1}{2}\mathbf{a}) - (\mathbf{a} + \frac{1}{2}\mathbf{c})$

6 $= \frac{1}{2}\mathbf{c} - \frac{1}{2}\mathbf{a} = \frac{1}{2}(\mathbf{c} - \mathbf{a})$

7 $AC = OC - OA = \mathbf{c} - \mathbf{a}$, so $MN = \frac{1}{2}AC$.

8 MN is a multiple of AC, so they are parallel, and MN is half the length of AC (ratio 1 : 2).

Answer: (a) $\mathbf{a} + \frac{1}{2}\mathbf{c}$ (b) $\frac{1}{2}\mathbf{a} + \mathbf{c}$ (c) $MN = \frac{1}{2}AC$, so parallel with lengths in the ratio 1 : 2